Kubernetes & Pods - Complete Beginner Guide

1. What is Kubernetes?

Kubernetes (K8s) is an open-source container orchestration platform that automates deployment, scaling, networking, and management of containerized applications.

2. Why Kubernetes?

• Automatically restarts failed containers.

• Scales applications up or down.

• Balances traffic between Pods.

• Performs rolling updates and rollbacks.

• Manages storage and networking.

3. Kubernetes Architecture

Control Plane: Manages the cluster.

API Server: Receives kubectl requests.

Scheduler: Assigns Pods to Nodes.

Controller Manager: Maintains desired state.

etcd: Stores cluster data.

Worker Node: Runs application Pods.

Kubelet: Communicates with control plane.

Container Runtime: Runs containers.

4. What is a Pod?

A Pod is the smallest deployable unit in Kubernetes. A Pod contains one or more tightly coupled containers sharing the same network and storage.

5. Pod Lifecycle

• Pending

• Running

• Succeeded

• Failed

• Unknown

• CrashLoopBackOff (restart failure)

6. kubectl Commands

kubectl version
Shows client/server version.

kubectl cluster-info
Displays cluster information.

kubectl get nodes
Lists worker/control-plane nodes.

kubectl get pods
Lists Pods in default namespace.

kubectl get pods -A
Lists Pods in all namespaces.

kubectl get pods -o wide
Shows Pod IP and Node.

kubectl describe pod
Detailed Pod information and events.

kubectl logs
Shows container logs.

kubectl exec -it -- /bin/bash
Opens shell inside container.

kubectl delete pod
Deletes a Pod.

kubectl apply -f pod.yaml
Creates/updates resource from YAML.

kubectl delete -f pod.yaml
Deletes resource defined in YAML.

7. Creating a Pod

Method 1: Command

kubectl run nginx-pod --image=nginx
Creates a Pod named nginx-pod using the nginx image.

Verify

kubectl get pods

kubectl describe pod nginx-pod

kubectl logs nginx-pod

Delete

kubectl delete pod nginx-pod

8. Creating a Pod using YAML

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx
    image: nginx:latest
    ports:
    - containerPort: 80
```

Create: kubectl apply -f pod.yaml

Delete: kubectl delete -f pod.yaml

9. Useful Troubleshooting

kubectl get events - Shows cluster events.

kubectl describe pod - Find scheduling/image errors.

kubectl logs - Application logs.

kubectl exec -it -- sh - Debug inside container.

10. Best Practices

• Use YAML instead of kubectl run for production.

• Set resource requests and limits.

• Use labels for organization.

• Use Deployments instead of standalone Pods for production.

• Monitor Pod status regularly.